

Semantic Drift Analysis Over Time

Tracing Linguistic Evolution via Diachronic Word Embeddings

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NLP Project

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Introduction & Objective

- **Objective:** To track, analyze, and visualize how the meaning of words, phrases, and entire sentences evolve over two centuries (1800-1990).
- **The Phenomenon:** Language is not static; semantic drift occurs continuously due to cultural, technological, and societal transformations.
- **Approach:** Leveraging diachronic historical word embeddings to capture and visualize changes in contextual relationships.

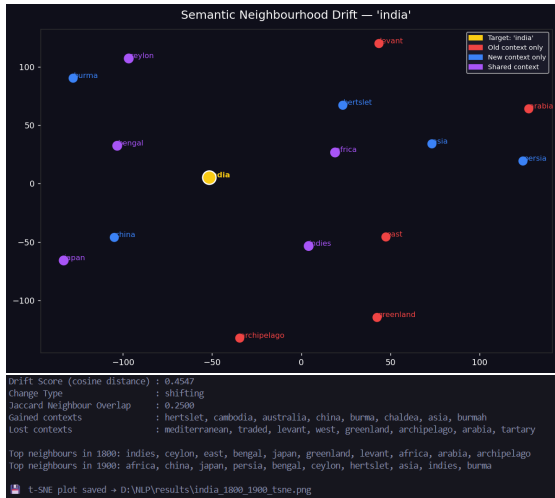
1. Previous Research & Foundation

- **Kim et al. (2014):** Introduced continuous representations for diachronic semantic tracking using Neural Language Models.
- **Hamilton et al. (2016):** Established the Orthogonal Procrustes alignment technique for mapping historical embeddings from distinct time periods into a shared coordinate space.
- **Lee et al. (EMNLP 2025):** Shifted the paradigm from "Detection" (scoring a specific known word) to "Discovery" (ranking all words in a dictionary by meaning shift).
- Most of the previous research focuses heavily on tracing isolated, individual words rather than complete contextual sentences.

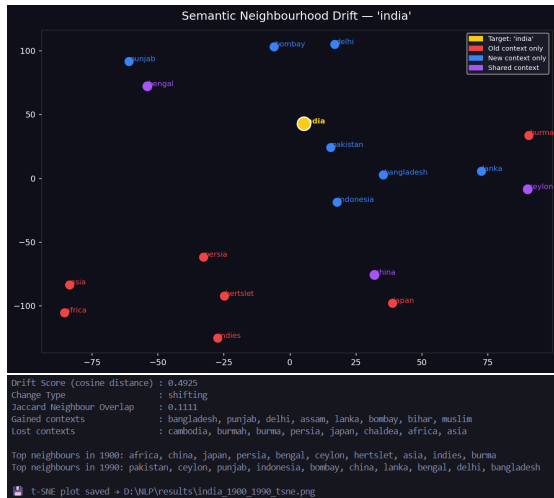
2. Technical Implementation

- **Dataset:** Stanford / Google N-gram Historical Word Vectors (SGNS format).
- **Temporal Space Alignment:** Orthogonal Procrustes was used on high-dimensional vectors.
- **Backend Architecture:** Python, Flask, NumPy, and Scikit-learn. We utilized memory-mapped matrices (`mmap`) to efficiently query gigabytes of embedding data in real-time.
- **Frontend Visualizations:** Vanilla CSS glassmorphism UI, Plotly.js for interactive mapping, and Matplotlib/Seaborn for generating downloadable static heatmaps.

3. Analysis Module: Diachronic Timeline



1800–1900 t-SNE

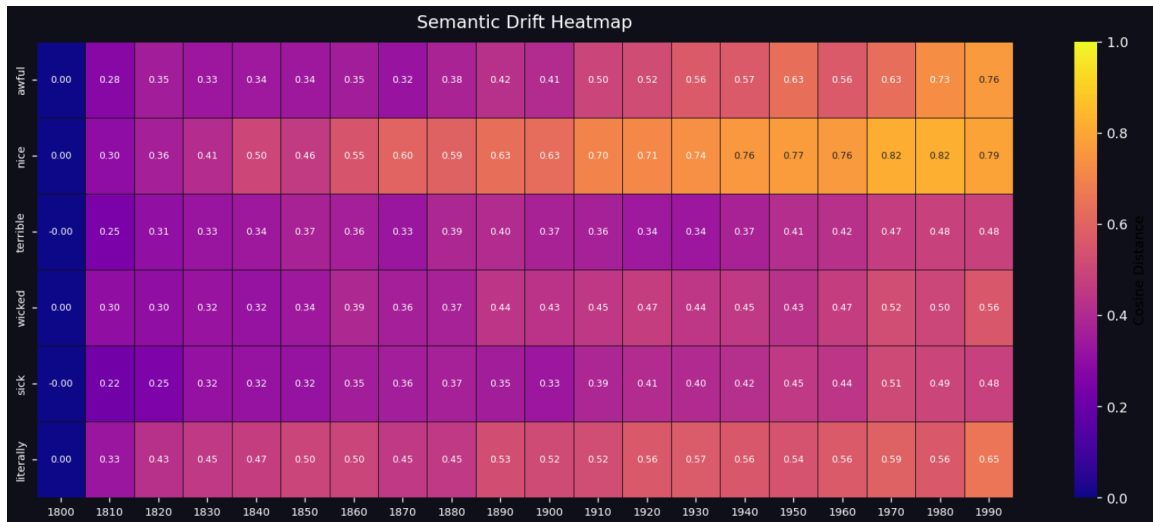


1900–1990 t-SNE

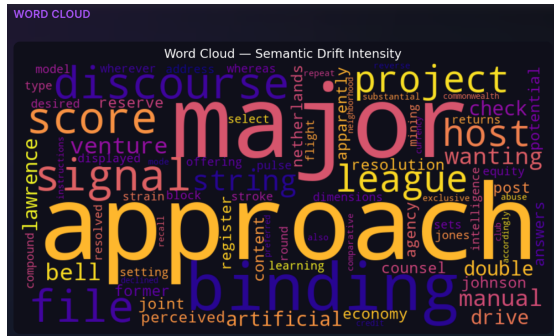
3. Analysis Module: Correlation Heatmaps



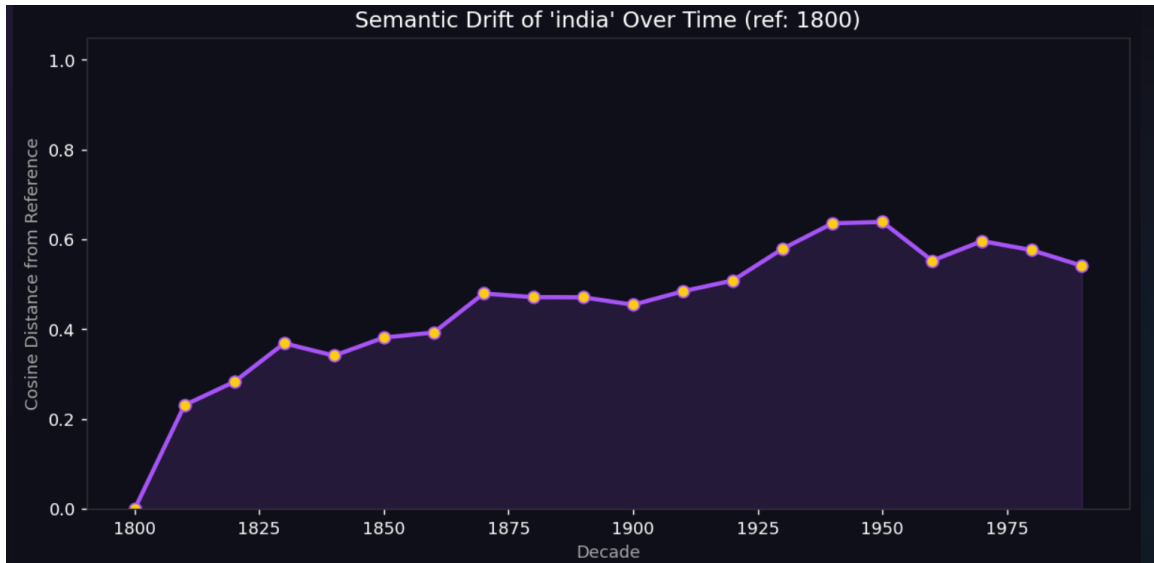
3. Analysis Module: Word Semantic drift Heatmaps



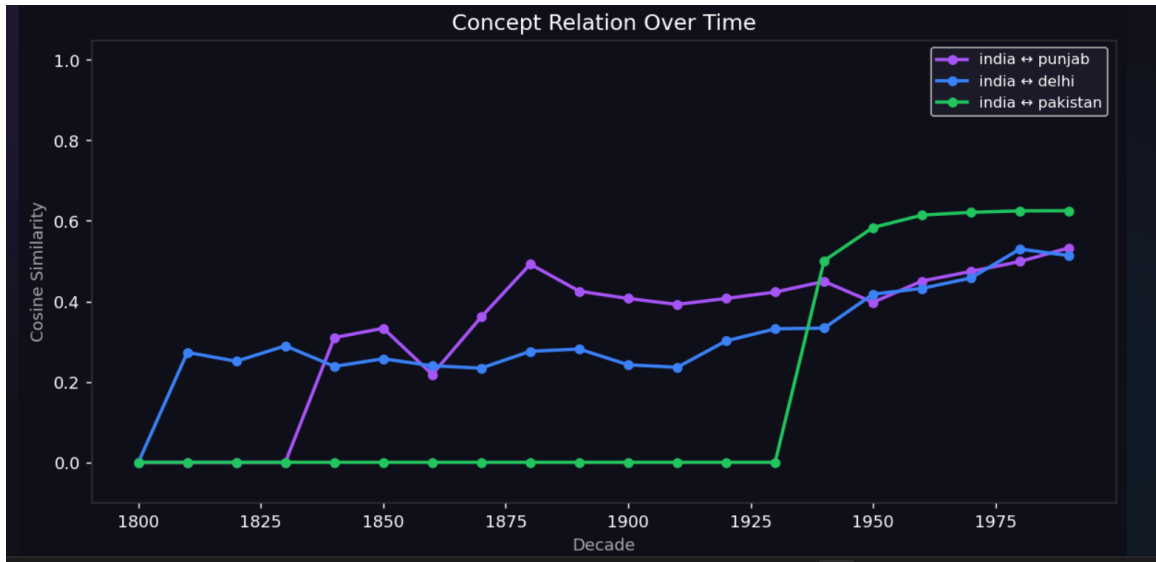
3. Analysis Module: Top Drifted Vocabulary



3. Analysis Module: One words sematic drift



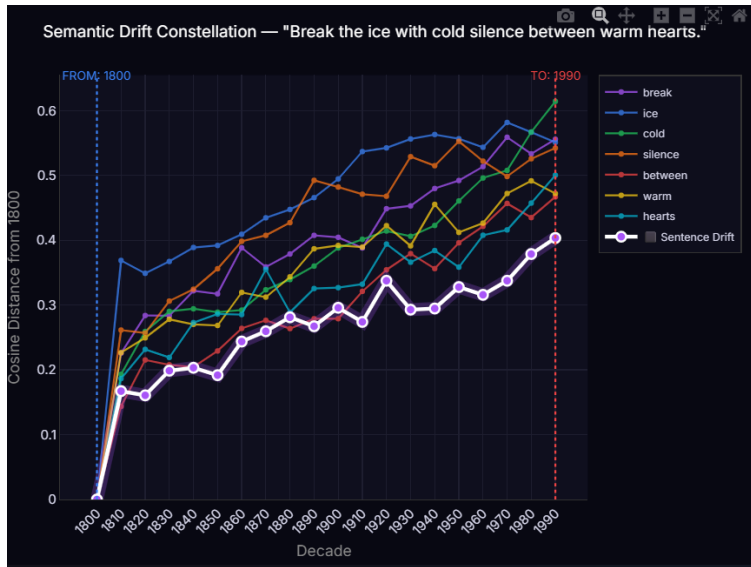
3. Analysis Module: Two words cosine similarity



4. Innovation: Sentential Discovery & Logic

- **Counterfactual Contribution Scoring:** I score how much a specific word *drives* the overall meaning shift of the sentence, mathematically bridging word decay and sentence decay.
- **Domain Transitions:** I implemented an automated domain mapping algorithm based on nearest-neighbor lexical overlap to interpret category jumps (e.g., Agriculture → Technology).
- No existing open-source tool visualizes the dynamic restructuring of semantic bonds *within* an idiomatic phrase over time. It effectively elevates the field from observing individual words to evaluating contextual shifts.

4. Innovation: The Drift Constellation



- **Sentence-Level Tracking:**
While prior tools analyze words one by one, the platform tracks the evolutionary trajectories of all constituent words within a complete sentence simultaneously.

- Developed a scalable, interactive application capable of serving large datasets for historical linguists and NLP researchers.
- Successfully extended the latest top-K discovery frameworks to multi-variable sentential representations.
- **Future Work:** Extend capabilities to newer historical ranges (2000-2024), deploy Contextualized Embeddings via transformer architectures (Temporal BERT), and implement multi-lingual support.

Thank You

Any questions?